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Google Scholar as a Resource for Systematic Reviews in Clinical Medicine

Matthew E. Falagas[1,2,3] | Paraskevi Maria Paliogianni[1] | Dimitrios S. Kontogiannis[1] | Dimitrios Ragias[1] | Elizabeth Johnson[2]

[1]Alfa Institute of Biomedical Sciences, Athens, Greece | [2]School of Medicine, European University Cyprus, Nicosia, Cyprus | [3]Department of Medicine, Tufts University School of Medicine, Boston, Massachusetts, USA

Correspondence: Matthew E. Falagas (m.falagas@aibs.gr)

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ABSTRACT

Background: Authors of systematic reviews must select, among several options, the databases for searching articles for inclusion in their analyses. Google Scholar is readily available, easy to use, and widely accepted for everyday information searches, including scientific research. However, there is no consensus for its use as a resource in systematic reviews.

Methods: This study assessed the proportion of systematic reviews that used Google Scholar as a resource, the search strategies used, and the number of potentially missed articles if the search in Google Scholar was omitted and focused on PubMed, Cochrane Library, and Scopus. We also analyzed data from the most recent systematic reviews in clinical medicine indexed in PubMed that listed Google Scholar as one of the resources used for literature searches.

Results: The term 'Google Scholar' was included in the title and/or abstract of 6.1% of systematic reviews archived by PubMed, compared to 37.5%, 36.1%, 31.8%, 18.5%, and 14.1% for the terms 'PubMed', 'Embase', 'Cochrane', 'Web of Science', and 'Scopus', respectively. Almost all (1029/1030) articles in the results section of the evaluated systematic reviews could be found in Google Scholar searches. If Google Scholar was omitted as a resource, the missed articles were 5% (53/1029). Twenty-one of 50 (42%) of the evaluated systematic reviews did not mention the number of articles identified from Google Scholar searches.

Conclusion: Google Scholar, as the most inclusive resource, should be used along with other established resources for systematic reviews. Advances in artificial intelligence may facilitate its use for this scientific purpose.

1 | Introduction

A systematic review employs well-defined scientific methods to identify, screen, select, and synthesize relevant published research data on a defined question [1, 2]. When a statistical synthesis is appropriate, a meta-analysis enables the combination of findings from multiple relevant original studies. This provides for more robust conclusions, in the absence of considerable clinical and statistical heterogeneity [1, 2]. Systematic reviews and meta-analyses potentially offer professionals the necessary data to make evidence-based decisions [1, 3]. They also help expert committees of medical societies generate guidelines, playing an important role in modern clinical medicine [1, 3].

Google Scholar, a subset of the broader Google search engine, facilitates retrieving a wide range of publications, including full-text journal articles, conference presentations, technical reports, preprints, theses, and books, as well as other types of publications. Additionally, it offers access to a selection of worldwide web pages that are considered academic [4]. Several of these items are open-access with free retrieval options [5, 6]. Google Scholar is particularly valuable in citation linking and accessing 'gray literature', such as conference presentations, government

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and industry reports, and other publications not typically published by traditional publishing companies and organizations, sources that are often not indexed in traditional academic databases [7].

Google Scholar is quick and straightforward to search, much like the main Google search engine index. Using a relevancy algorithm that is well-guarded by the owning company, the search results are displayed after retrieving document or page matches, depending on the searched keywords. Articles with large numbers of citations are usually listed on the first pages of the search results, making Google Scholar effective in identifying highly cited, and hence influential, articles [8]. An important feature is Google Scholar's 'cited by' function, which simplifies locating relevant citations by connecting a retrieved result to other resource entries that mention it [4]. Despite its availability as a free resource, there is no consensus for its use in systematic reviews in clinical medicine and other scientific fields.

This study assessed the suitability of Google Scholar as a resource for systematic reviews. Specifically, it reviewed the frequency with which it is used for systematic reviews, as well as what researchers would potentially miss if it were omitted as an evidence-collecting tool in their execution. A central question addressed in this study is whether a universal search engine like Google Scholar could serve as a reliable tool for authors of systematic reviews, as there is controversy regarding Google Scholar's consistent reproducibility. In light of the current immense overload of scientific information, the question arises whether a simple search engine like Google Scholar adheres sufficiently to the robust scientific methodology required for sound systematic reviews.

2 | Methods

2.1 | Study Objectives

This study aimed to assess first the proportion of systematic reviews that used Google Scholar as a resource for identifying relevant articles. Second, the proportion of systematic reviews that reported the number of identified articles from Google Scholar search separately from those found using other resources. Third, the specific search strategies used in searching Google Scholar. Fourth, the proportion of articles that could be retrieved only from searching Google Scholar (that would be missed from identification if the search focused only on PubMed, Cochrane Library, and Scopus but did not include Google Scholar as a potential resource).

2.2 | Frequency of Use of Google Scholar as a Resource

To estimate the frequency of systematic reviews that used Google Scholar, we assessed the total number of systematic reviews retrieved by PubMed (on 17 February 2025) that included Google Scholar as a source. This was achieved using the search strings: 'systematic review', 'systematic review' AND ('Google'[Title/Abstract]), and 'systematic review' AND ('Google Scholar'[Title/Abstract]).

Then, we used the same search strings by applying the filter [Title] for 'Google' and 'Google Scholar' and the results were subtracted from the ones using the previously mentioned search strategies (without the use of the filter) to calculate the number of remaining articles that mentioned 'Google' or 'Google Scholar' only in their abstract. The decision to include both the terms 'Google' and 'Google Scholar' arises from the fact that some authors could use the term 'Google' instead of 'Google Scholar' in their search strategy. We followed the same search strings by applying separately the terms 'PubMed', 'Embase', 'Cochrane', 'Web of Science', and 'Scopus' (instead of 'Google Scholar') to compare the results for 'Google' and 'Google Scholar' with other frequently used resources for systematic reviews.

2.3 | Selection of a Sample of Analyzed Systematic Reviews

To address the remaining study objectives reported above (the last three of four objectives), we extracted and analyzed data from the 50 most recent systematic reviews in clinical medicine retrieved from PubMed (the search was performed on December 27, 2024). Systematic reviews from other potentially relevant biomedical fields, such as informatics, dentistry, psychology, or biochemistry, were excluded from our analysis. We applied the filters 'English' and 'humans' in our search.

2.4 | Report on the Number of Identified Articles From Google Scholar

We examined the search strategies reported by the authors of each evaluated systematic review by examining the flow chart (identification, screening, and selection of articles) and the full text. Then, the 50 systematic reviews included in our analysis were divided into 3 groups: those that reported in their flow chart and/or full text the total number of identified articles (only as a sum from all resources), those that reported the number of articles retrieved from each resource separately (including Googe Scholar), and those that reported separately the number of articles retrieved from other resources, but not from Google Scholar. We also examined whether the term 'Google Scholar' was mentioned only in the abstract and not in the full text.

2.5 | Search Strategy of Google Scholar as a Resource

We extracted data from the flowcharts, full texts, and supplementary materials of each evaluated systematic review, as well as the search strings used by the authors in their Google Scholar search strategy, if applicable. Also, we examined whether relevant details were provided (e.g., the 'all-in-title' filter).

2.6 | Missed Articles if Google Scholar Is not Used as a Resource

To determine potential missed articles by omitting Google Scholar as a resource, we assessed whether the individual articles included in our evaluated sample of systematic reviews in

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clinical medicine were identifiable by separate searches on the following databases: PubMed, Cochrane Library, and Scopus. We adapted the search strategy to the characteristics and search options of resources. More specifically, for Google Scholar and PubMed, we used the title of each article to find it in the respective resources. For Cochrane Library and Scopus, we used the digital object identifier (DOI) of each article, if available; in cases where the DOI was not available, the title of the article was searched, by applying the filters 'record title' and 'article title' in Cochrane Library and Scopus, respectively.

3 | Results

3.1 | Frequency of Use of Google Scholar as a Resource

Table 1 presents the frequency of use of Google Scholar and other resources in systematic reviews, by implementing different search strategies in PubMed. The search for 'systematic review' retrieved 382,276 results. The subset of systematic reviews that included the term 'Google' in the title and/or abstract was 24,369/380,925 (6.4%), and for the term 'Google Scholar' in the title and/or abstract was 23,097/380,925 (6.1%). In comparison, the subset of systematic reviews that included the terms 'PubMed', 'Embase', 'Cochrane', 'Web of Science', and 'Scopus' in the title and/or abstract, were 142,673/380,925 (37.5%), 137,695/380,925 (36.1%), 121,311/380,925 (31.8%), 70,282/380,925 (18.5%), and 54,066/380,925 (14.1%), respectively.

3.2 | Reporting of the Number of Identified Articles From Google Scholar

Table 2 presents the characteristics of the reporting of the use of Google Scholar in the 50 evaluated systematic reviews and the proportion of articles missed if the Google Scholar search was omitted. All 50 systematic reviews included in our analysis

TABLE 1 | Frequency of the terms 'Google', 'Google Scholar', 'PubMed', 'Embase', 'Cochrane', 'Web of Science', and 'Scopus' employed in the title and/or abstract of systematic reviews retrieved by PubMed ($N = 380{,}925$)[a].

Searched terms	Searched term in the title or abstract[b]	Searched term only in the title[b]	Searched term only in the abstract[c]
		***n* (%)**	
'Google'	24,369 (6.4)	33 (0.009)	24,336 (6.4)
'Google Scholar'	23,097 (6.1)	9 (0.002)	23,088 (6.1)
'PubMed'	142,673 (37.5)	56 (0.014)	142,617 (37.4)
'Embase'	137,695 (36.1)	17 (0.004)	137,678 (36.1)
'Cochrane'	121,311 (31.8)	1110 (0.291)	120,201 (31.6)
'Web of Science'	70,282 (18.5)	12 (0.003)	70,270 (18.4)
'Scopus'	54,066 (14.1)	11 (0.003)	54,055 (14.2)

[a]The search was performed on 17/February/2025.
[b]Searches were performed by using the filters [Title/Abstract] and [Title] for searched terms in the title or abstract (second column) and title (third column), respectively.
[c]Subtraction of the number of the third column from the number of the second column in each row (fourth column).

TABLE 2 | Characteristics of the use of Google Scholar in systematic reviews and missed articles if Google Scholar search is omitted.

Study objective	Systematic reviews *n*/*N* (%)
Report of the number of articles retrieved from the searched resources (of the sample of 50 evaluated systematic reviews)	
Report of the total number of retrieved articles only as a sum (from all searched resources)	14/50 (28)
Separate report of the number of articles retrieved from other searched resources (but not from Google Scholar)	7/50 (14)
Separate report of the number from each searched resource	29/50 (58)
Search strategy for Google Scholar (when reported in the sample of 50 evaluated systematic reviews)	
Use of the 'all-in-title' filter	4/50 (8)
Explicit report of search strings used for Google Scholar	9/50 (18)
	Articles *n*/*N* (%)
Missed articles if the Google Scholar search is omitted (of the 1030 articles included in the sample of 50 evaluated systematic reviews)	
Articles missed if the Google Scholar search is omitted (searches performed only in PubMed, Cochrane Library, and Scopus)	53/1029 (5)

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reported the total number of identified publications as a sum of articles retrieved from all databases searched. From those, 14/50 (28%) reported in their flow chart and/or full text the total number of identified articles (for subsequent screening and selection), only as a sum from all the resources they searched, including Google Scholar, without reporting separately the number of articles they retrieved from each resource. Additionally, 7/50 (14%) reported separately the number of articles retrieved from each resource searched, excluding Google Scholar. Twenty-nine of 50 (58%) of the evaluated reviews reported the number of articles retrieved from each database (including Google Scholar) separately. In addition, 3/50 (6%) of the evaluated reviews mentioned Google Scholar as a resource only in the abstract, but not in the methods section.

3.3 | Search Strategy of Google Scholar as a Resource

Among the 50 systematic reviews, 4/50 (8%) specified that their search in Google Scholar was done only with the 'all-in-title' filter, which is applied through the advanced search option and searches the terms only in the title of the articles (Table 2). Also, the search strings or terms used in Google Scholar searches were explicitly reported in 9/50 (18%) of the evaluated systematic reviews.

3.4 | Missed Articles Due to the Omission of Google Scholar as a Resource

The fourth objective of our analysis was to calculate the proportion of articles included in the evaluated sample of systematic reviews by our team that would be missed if Google Scholar were not included in the search process, in addition to other databases. In two of the 50 evaluated systematic reviews [9, 10], 13 articles (12 and 1 articles from each review) included in their results section could not be found in the studied resources, including Google Scholar. These 13 articles were excluded from further analysis. Additionally, in another evaluated systematic review [11], two articles included in the tables of the results section reported the first author and date of publication. However, they were not included in the reference list; these two articles were also excluded from further analysis. The remaining 1030 articles in the results section of the 50 evaluated systematic reviews were assessed to determine whether they could be found in 4 different resources (Google Scholar, PubMed, Cochrane Library, and Scopus). One thousand and twenty-nine of 1030 (99.9%) articles could be found in Google Scholar searches (the single article not found in Google Scholar could be found in Google and Scopus searches).

In 33/50 (66%) of the evaluated systematic reviews, none of the articles included in their results section would have been missed by omitting the search of Google Scholar, because all relevant articles could be retrieved by one or more of the other resources (PubMed, Cochrane Library, and Scopus). In the remaining 17/50 (34%) of the evaluated systematic reviews, some articles could only be retrieved by Google Scholar, but not by any other resource (PubMed, Cochrane Library, and Scopus). More specifically, in 9/17 (53%) and 4/17 (24%) of these reviews, one and two of their articles could only be retrieved by Google Scholar, respectively (these numbers were 3%–14% and 25%–33% of the total number of their articles, respectively). In the remaining 4/17 (24%) of the systematic reviews, the articles that could only be retrieved by Google Scholar of the total number of their articles were 5/85 (6%), 4/18 (22%), 15/65 (23%), and 12/33 (36%).

In total, 53/1029 (5%) of articles included in the evaluated systematic reviews would have been missed by omitting the search of Google Scholar (and performing searches only in PubMed, Cochrane Library, and Scopus) (Table 2). Concerning these 53 articles, 2 of them were conference abstracts. The majority [27 (53%)] from the rest 51 of these articles were published in journals originating from Asia (India 10, Pakistan 6, Iran 4, Iraq 4, China 1, Korea 1, Turkey 1); 9 (18%) from North America (USA); 8 (16%) from Africa (Egypt 4, Ethiopia 2, South Africa 1, Uganda 1); 6 (12%) from Europe (UK 3, Greece 1, North Macedonia 1, Poland 1); and 1 from South America (Brazil). Forty-six of these 51 articles had available citation counts in Google Scholar, ranging from 1 to 29 (with a median of 2) citations per article. The median year of their publication was 2017.

4 | Discussion

This study investigated the use of Google Scholar as a resource for the literature searches in systematic reviews. The data suggest that Google Scholar is used less frequently than other article resources (PubMed, Embase, Cochrane Library, Web of Science, and Scopus) for this type of publication. The term 'Google Scholar' was included in the title and/or abstract of 6.1% of systematic reviews archived by PubMed, compared to 37.5%, 36.1%, 31.8%, 18.5%, and 14.1% for the terms 'PubMed', 'Embase', 'Cochrane', 'Web of Science', and 'Scopus', respectively. Although we focused only on the title and/or abstract, and thus, higher proportions of systematic reviews may report the use of all these resources in their methods section. Nonetheless, the available data clearly indicates the infrequent use of Google Scholar compared to other resources.

A considerable proportion (42%) of the study sample did not report the number of articles retrieved from Google Scholar. Most reported the total number of articles only as a sum from all searched resources, raising concerns about the actual extent of Google Scholar's use as a resource. Furthermore, 8% of the systematic reviews included in this analysis used the 'all-in-title' filter for their searches in Google Scholar, limiting the scope of the search compared to broader keyword searches that include not only in the title but also in other fields, including the abstract.

According to the results of this study, if only PubMed, Cochrane Library, and Scopus were used in a systematic review without Google Scholar, about 5% of relevant articles would be missed. Although these missed articles had few citations in Google Scholar and half were published in Asian journals, their contribution to a robust evidence synthesis cannot be ignored.

In our analysis of articles included in the evaluated systematic reviews, we selected PubMed, the Cochrane Library, and Scopus over other databases such as Embase and Web of Science for several reasons. PubMed is a comprehensive database

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for biomedical sciences, available free of charge. Cochrane Library focuses on evidence-based articles, including randomized controlled trials and systematic reviews. Scopus is readily available, affordable, and offers wide coverage across scientific fields, making it an appropriate choice for evaluations that require multi-dimensional observations. It also performs well in citation analysis and is simpler for searches compared to other resources that rely on somewhat complicated entry terms, such as Embase, where Emtree terms should be applied for search refinement.

Previous publications have assessed various aspects of using Google Scholar as a resource for systematic reviews [12]. After recreating the search strategy of 10 meta-analyses in the field of otolaryngology, it was demonstrated that by screening the first 500 results in Google Scholar (sorted by relevance), 99% of relevant articles included in these meta-analyses were identified [13]. Also, using Google Scholar in combination with different databases (such as Embase, Web of Science, and MEDLINE) increased the overall recall (defined as the proportion of relevant articles retrieved to the total number of relevant articles that existed in those databases) [14].

Google Scholar is a valuable search engine that is useful for searches for various purposes, including scientific research [15, 16] and answering everyday clinical questions [17]. Moreover, it has high coverage [18–20] and more citation counts [21, 22] of articles compared to other resources. However, Google Scholar has limitations when used for systematic reviews [23]. For example, one can only access up to 1000 results, even if the results exceed 1000. Also, there is a 256-character limit on the search strings [24]. In addition, it offers limited search options, even in its advanced search feature. For example, it is impossible to use combinations of disjunction and conjunction of phrases using the advanced search strategy of Google Scholar. Instead, one must create the search string in the main search option bar (not the advanced search option) and manually insert parentheses and Boolean operators wherever needed, a process that may be considered complex by some users [24].

There are also issues related to the reproducibility, accuracy, and recall of the results of Google Scholar searches [14, 20, 24–28]. Reproducibility guarantees that identical searches get consistent results, independent of location or time. Systematic evaluations must provide reliability and transparency. Variations that cannot be clarified by database updates, such as those resulting from institutional access differences or VPN usage, suggest possible bias and compromise search integrity [27]. Google Scholar's reproducibility is sometimes inconsistent, with search results changing unpredictably between repeated searches, regardless of natural database growth [27]. However, recall and precision depend on the experience of the user;

TABLE 3 | The main advantages and disadvantages of Google Scholar as a resource for various scientific purposes, including systematic reviews.

Characteristic	Advantages	Disadvantages
Search strategy	'AND', 'OR', NOT', and 'AROUND (n)' operators available	Wildcards and truncation operators are unavailable Limit of 256 characters for each search string
Access to literature	Available links for the free identification of articles Off-campus access (access to subscribed journals even while traveling or at home)	
Notifications	Receive e-mails with new results from a specific search through the 'create alert' option	
Organization	Organize articles by adding them to 'my library' and using labels	Unavailable to bulk export all the results of a search in a citation manager software Search history unavailable
Coverage of articles	Extended coverage of publications (including 'gray literature')	
Identification of relevant articles	Search for relevant articles of a specific study using the 'related articles' option	Only the first 1000 results are available for retrieval Only 10 or 20 results are available per page May give different results for the search of the same terms with a different order [e.g. search of ('term A' OR 'term B') and ('term B' OR 'term A') may give different numbers of results]
Updates of the resource	Frequently updating for new articles, several times per week	For an already existing article in Google Scholar, modifications and updates take 6 to 9 months (even a year) to be implemented
Filters		Limited filters; only date, journal, and author filters are available

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experienced librarians and clinicians achieve better results than non-experienced users [29, 30]. Either way, the use of Google Scholar in systematic reviews, in addition to other resources (3 or more for a comprehensive systematic review), increases the overall recall of the search strategy. More articles can be identified if Google Scholar is added as a resource.

Taken collectively, the advantages and disadvantages, along with the results of our study, suggest that Google Scholar cannot be used as a sole resource for systematic reviews, similar to other resources, including PubMed, Embase, Cochrane Library, Web of Science, and Scopus. However, the results suggest that it could be a valuable supplementary resource in combination with other traditional resources, especially for publications and reports of the 'gray literature', to enhance the identification, screening, and selection of relevant articles in systematic reviews. Our data suggest that Google Scholar is the most inclusive resource for systematic reviews. Also, it offers an extensive citation linking feature, which is particularly helpful for scientific purposes. Table 3 presents a summary of the advantages and disadvantages of Google Scholar as a resource for various scientific purposes, including systematic reviews.

Artificial intelligence platforms have recently been utilized frequently and could facilitate and automate the screening process for articles in systematic reviews [31]. Given that Google Scholar is more inclusive than other available resources for scientific research (such as PubMed, Embase, Cochrane Library, Scopus, Web of Science, and others), a reliable automated system that minimizes the time required for article screening will benefit researchers. In addition, some suggest that, if Google Scholar is used as a resource of gray literature in a study, then the authors should focus their searches after page 20 of the results [7]. Others suggest that if searching the first 200 results of Google Scholar in combination with citation tracking of relevant articles, then the coverage is 98% [13]. There is still no consensus on the recommendations for the optimal use of Google Scholar.

Our study is not without limitations. Our data on the frequency of use of Google Scholar as a resource for systematic reviews archived by PubMed was based on the term 'Google Scholar' in the title and/or abstract (and not in the methods section). This potentially leads to considerably underestimated estimates of the use of all resources; however, the relative use offers useful insights. In addition, the term 'Google Scholar' in the abstract of a systematic review does not necessarily mean that Google Scholar was used as a resource (in an extreme case, a review could even mention in the abstract that Google Scholar was not used as a resource).

Also, in our analysis, we used PubMed to identify a sample of evaluated systematic reviews that included the term 'Google Scholar' in the title and/or the abstract, but not in all fields (e.g., MeSH terms, MeSH subheadings, and text words). Thus, systematic reviews included in other databases (such as Embase), but not in PubMed, as well as systematic reviews that mention Google Scholar in other fields other than title and/or abstract in PubMed, could not be retrieved. However, in an exploratory analysis, we examined whether searches for systematic reviews in PubMed using, in addition to the term 'systematic review', the term 'Google Scholar' in the [Title/Abstract] field versus the [all] or [Text Word] field yields different results (the searches were done on 06/June/2025). The yield of the searches (number of articles) was precisely the same (24,487 articles) using the term 'Google Scholar' in the [Title/Abstract] field versus the [all] or [Text Word] field. Also, the yield was almost the same with the term 'Google' in the [Title/Abstract] field versus the [all] or [Text Word] field, specifically, there were 43 additional articles, i.e., 0.17% more using the [all] field (25,823 vs. 25,780 articles).

The proportion of articles that could be retrieved only from searching Google Scholar (and thus, that would be missed from identification if the search included PubMed, Cochrane Library, and Scopus) was the most crucial outcome in our analysis. The estimates for this outcome are not expected to be systematically different in samples of systematic reviews with various selection strategies (beyond the effect of randomness), i.e., if the term 'Google Scholar' appears or not in the title or abstract. However, a limitation of our article is that we did not expand our analysis to a different study sample of systematic reviews that did not include the term 'Google Scholar' in the title or abstract.

In our analysis of missed articles, if Google Scholar was omitted as a resource, there were outliers. Specifically, in two of the evaluated systematic reviews, a large proportion of articles were exclusively found in Google Scholar. This observation suggests that, for the majority of systematic reviews, the true proportion of missed articles if Google Scholar is omitted as a resource may be considerably lower than 5%, the average estimate we reported. Indeed, if the outliers were excluded from our analysis, this proportion would be 3%. Also, the sample of 50 systematic reviews included in parts of our analysis may not be large enough to draw firm conclusions. Finally, we focused on clinical medicine, thus, the results from our study may not be generalizable to systematic reviews in other scientific fields.

5 | Conclusion

Our findings suggest that Google Scholar, although the most inclusive available resource, is considerably less frequently used for the identification of articles for inclusion in systematic reviews in clinical medicine, compared to other resources, namely PubMed, Embase, Cochrane Library, Web of Science, and Scopus. Google Scholar should be used, along with at least three other established resources, for comprehensive systematic reviews. Advances in artificial intelligence may facilitate its use for this scientific purpose.

Author Contributions

Matthew E. Falagas had the idea for the article. All authors contributed to the methodology used in the article. SP and Dimitrios S. Kontogiannis did the literature search, data extraction, and tabulation. Matthew E. Falagas, SP, and Dimitrios S. Kontogiannis contributed to the first version of the manuscript. Dimitrios Ragias and Elizabeth Johnson revised the manuscript. All authors contributed to subsequent revisions and agreed with the final version of the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data used in the conduct of this study are available upon request.

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